

# Haozhe Lei

Ph.D. Candidate, Electrical and Computer Engineering | NYU WIRELESS

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## RESEARCH PROFILE

**RF sensing and probabilistic spatial intelligence for autonomous systems.**

- **Core problem and inference:** I treat wireless localization as belief estimation, not point prediction: one multipath snapshot can support several transmitter positions and headings when antennas are directional or maps and digital twins are mismatched. I learn candidate-wise likelihoods, form calibrated spatial posteriors, and fuse observations.
- **World models and decisions:** Ray-tracing wireless digital twins provide imperfect structured priors; I align them to measurements through real-to-sim calibration and learn map- and pose-conditioned neural scores. Physics-informed sequential decisions connect RF evidence to navigation under partial observability.
- **System translation:** The program spans RF localization, robotic sensing and navigation, and validation on a 10-GHz FR3 platform. Complementary directions include UE-centric multi-band adaptation and vision-language-guided object-centric RGB-D spatial memory for embodied agents.

## EDUCATION

**New York University**

Ph.D. in Electrical and Computer Engineering

Advisor: Prof. Sundeep Rangan, Director, NYU WIRELESS

Sept. 2022 – Expected May 2027

Fellowship: Ernst Weber Fellowship, 2023

**New York University**

M.S. in Computer Engineering

Sept. 2020 – May 2022

**China Agricultural University**

B.E. in Electrical Engineering and Automation

Sept. 2015 – Jun. 2019

## RESEARCH EXPERIENCE

**NYU WIRELESS, New York University**

Graduate Research Assistant; Advisor: Prof. Sundeep Rangan

Brooklyn, NY, USA

Sept. 2022 – Present

- **Probabilistic RF localization and measurement-aligned wireless digital twins.** Lead end-to-end development of likelihood-based and neural posterior localization under sparse measurements and digital-twin mismatch, from problem formulation and algorithm design through implementation and evaluation. Methods include uncertainty calibration, Bayesian fusion, U-Net/attention spatial scoring, and measurement-aligned Sionna RT. Lead MAGNETAR from probabilistic formulation through RF-robot integration and experimental evaluation, combining multipath spatial posteriors with *MobiFR3* measurements for transmitter-pose inference. *Outputs:* IEEE GLOBECOM 2026; Asilomar 2026 (two papers); IEEE TVT manuscript under review; IEEE ICRA 2027 (two papers under review); IEEE TWC revision in preparation.
- **Wireless-aware robotic navigation: methods and experimental systems.** Lead *MobiFR3*, a heterogeneous 10-GHz FR3 robotic platform for localization experiments and SLAM-based navigation, integrating RF measurements with ROS 2 odometry, LiDAR/RGB sensing, and TurtleBot4 motion; Jackal integration is in progress. Direct system architecture, RF-robot synchronization, experiments, and evaluation while mentoring students in RF hardware operation and data collection. Earlier, co-developed physics-informed and program-guided reinforcement learning for zero-shot wireless indoor navigation. *Publications:* IEEE ICRA 2024; IEEE OJ-COMS 2025; and RLC IBRL Workshop 2025 (Spotlight).
- **UE-centric wireless digital twins for closed-loop multi-band adaptation.** Lead the research direction, problem formulation, digital-twin design, and learning methods for MCMB-HDT, spanning georeferenced propagation, multi-cell FR1/FR3 networks, handset arrays, and mobility. Develop Transformer-based sparse-history rate prediction and recurrent PPO for measurement-constrained array activation; mentor a junior researcher in model development, Sionna implementation, and evaluation. *Outputs:* IEEE ICC Workshops 2026; IEEE JSAC manuscript under review.

## PUBLICATIONS AND MANUSCRIPTS

Haozhe Lei is shown in bold; \* denotes equal contribution; † denotes corresponding author.

### Peer-Reviewed Publications

#### Wireless Sensing, Localization, and Robotic Systems

- [1] **Haozhe Lei**<sup>†</sup>, Roberto Bomfin, Marwa Chafii, and Sundeep Rangan. "LOCUS-DT: Localization via Observation-Conditioned Uncertainty Scoring with Digital Twins". *IEEE Global Communications Conference (GLOBECOM)*. 2026. [paper].
- [2] **Haozhe Lei**<sup>†</sup>, Hao Guo, Tommy Svensson, and Sundeep Rangan. "Beyond Point Estimates: Likelihood-Based Full-Posterior Wireless Localization". *Asilomar Conference on Signals, Systems, and Computers*. 2026. [paper].
- [3] Mingsheng Yin\*, Tao Li\*, **Haozhe Lei**<sup>\*</sup>, Yaqi Hu, Sundeep Rangan, and Quanyan Zhu. "Zero-Shot Wireless Indoor Navigation through Physics-Informed Reinforcement Learning". *IEEE International Conference on Robotics and Automation (ICRA)*. 2024. [paper].
- [4] Tao Li, **Haozhe Lei**, Hao Guo, Mingsheng Yin, Yaqi Hu, Quanyan Zhu, and Sundeep Rangan. "Digital Twin-Enhanced Wireless Indoor Navigation: Achieving Efficient Environment Sensing with Zero-Shot Reinforcement Learning". *IEEE Open Journal of the Communications Society* (2025). [paper].
- [5] Roberto Bomfin, **Haozhe Lei**, Sundeep Rangan, and Marwa Chafii. "Likelihood-Based Wireless Localization with Last-Bounce Spatial Features". *Asilomar Conference on Signals, Systems, and Computers*. 2026.
- [6] Tao Li, **Haozhe Lei**, Mingsheng Yin, and Yaqi Hu. "Reinforcement Learning with Physics-Informed Symbolic Program Priors for Zero-Shot Wireless Indoor Navigation". *Inductive Biases in Reinforcement Learning Workshop at RLC*. Spotlight paper. 2025. [paper].

#### Wireless Digital Twins and Multi-Band Systems

- [7] Ruibin Chen\*, **Haozhe Lei**<sup>\*†</sup>, Hao Guo, Marco Mezzavilla, Hitesh Poddar, Tomoki Yoshimura, and Sundeep Rangan. "Transformer-Based Rate Prediction for Multi-Band Cellular Handsets". *IEEE International Conference on Communications Workshops (ICC Workshops)*. 2026. [paper].

#### Additional Publications in Control, Transportation, and Security

- [8] Zhaoye Pan, **Haozhe Lei**, Fan Zuo, Zilin Bian, and Tao Li. "Distributed Online Convex Optimization with Nonseparable Costs and Constraints". *IEEE Control Systems Letters* 10 (2026), pp. 391–396. [paper].
- [9] Tao Li\*, Zilin Bian\*, **Haozhe Lei**<sup>\*</sup>, Fan Zuo\*, Ya-Ting Yang\*, Quanyan Zhu, Zhenning Li, Zhibin Chen, and Kaan Ozbay. "Digital Twin-Based Driver Risk-Aware Predictive Mobility Analytics for Real-Time Situational Awareness Through Cooperative Sensing". *IEEE Transactions on Intelligent Transportation Systems* (2025). [paper].
- [10] Ya-Ting Yang, **Haozhe Lei**, and Quanyan Zhu. "PRADA: Proactive Risk Assessment and Mitigation of Misinformed Demand Attacks on Navigational Route Recommendations". *IEEE Transactions on Information Forensics and Security* (2025). [paper].
- [11] Zilin Bian, Tao Li, **Haozhe Lei**, Fan Zuo, Ya-Ting Yang, Quanyan Zhu, Zhenning Li, and Kaan Ozbay. "Multi-level Traffic-Responsive Tilt Camera Surveillance through Predictive Correlated Online Learning". *Transportation Research Part C: Emerging Technologies* (2024). [paper].
- [12] **Haozhe Lei**<sup>\*†</sup>, Yunfei Ge\*, and Quanyan Zhu. "ADAPT: A Game-Theoretic and Neuro-Symbolic Framework for Automated Distributed Adaptive Penetration Testing". *IEEE Military Communications Conference (MILCOM)*. 2024. [paper].
- [13] Tao Li\*, **Haozhe Lei**<sup>\*</sup>, and Quanyan Zhu. "Self-Adaptive Driving in Nonstationary Environments through Conjectural Online Lookahead Adaptation". *IEEE International Conference on Robotics and Automation (ICRA)*. 2023. [paper].

## Manuscripts Under Review or in Revision

- [M1] **Haozhe Lei**<sup>†</sup>, Roberto Bomfin, Marwa Chafii, and Sundeep Rangan. “Site-Agnostic Posterior Inference for Indoor Localization with Ray-Tracing Wireless Digital Twins”. Revision in preparation for resubmission to IEEE Transactions on Wireless Communications. 2026.
- [M2] **Haozhe Lei**<sup>†</sup>, Hao Guo, and Sundeep Rangan. “Learning a Measurement-to-Posterior Map for Wireless Localization”. IEEE Transactions on Vehicular Technology, under review. 2026.
- [M3] **Haozhe Lei**<sup>†</sup>, Ruibin Chen, Yuhan Jiang, Ali Rasteh, Aditya Dhananjay, and Sundeep Rangan. “MAGNETAR: Multipath-Guided Spatial Posteriors for Transmitter Pose Inference in the Upper Mid-Band”. IEEE International Conference on Robotics and Automation (ICRA), under review. 2027. [\[paper\]](#).
- [M4] **Haozhe Lei**<sup>†</sup> and Sundeep Rangan. “MAPLE-RF: Efficient Probabilistic RF Source Localization in Partially Explored Environments”. IEEE International Conference on Robotics and Automation (ICRA), under review. 2027. [\[paper\]](#).
- [M5] Ruibin Chen<sup>\*</sup>, **Haozhe Lei**<sup>\*,†</sup>, Marco Mezzavilla, Hitesh Poddar, Tomoki Yoshimura, and Sundeep Rangan. “MCMB-HDT: A Multi-Cell Multi-Band Handset Digital Twin for Learning-Based Closed-Loop Array Activation”. IEEE Journal on Selected Areas in Communications, under review. 2026.

## RESEARCH MENTORING

### New York University

Feb. 2025 – Present

- **Ruibin Chen**, Ph.D. student

Feb. 2025 – Present

Mentor his research on *MCMB-HDT* and RF localization, including model development, Sionna implementation, and evaluation, within projects I lead. Joint work includes a peer-reviewed IEEE ICC Workshops 2026 paper and manuscripts under review at IEEE JSAC and IEEE ICRA 2027.

- **Yuhan Jiang**, M.S. student

Jul. 2026 – Present

Mentor research on RF hardware operation, calibration, and data acquisition for the FR3 robotic localization platform; co-author of *MAGNETAR*, under review at IEEE ICRA 2027.

- **Yuheng Liu and Ashesh Kaji**, M.S. students

2026 – Present

Mentor their joint project on object-centric RGB-D scene memory for embodied agents, including VLM-guided object proposals, depth-grounded 3D projection, cross-view association, and queryable scene representations for retrieval and localization. Yuheng joined in Jan. 2026; Ashesh joined in Jul. 2026.

## ADDITIONAL RESEARCH EXPERIENCE

### NYU C2SMARTER Center

Brooklyn, NY, USA

Graduate Research Assistant

Sept. 2023 – Mar. 2026

- Developed distributed online optimization, predictive camera control, and driver-risk-aware transportation digital twins for networked sensing and closed-loop urban mobility. Publications include IEEE Control Systems Letters (2026), IEEE T-ITS (2025), and *Transportation Research Part C* (2024).

### NYU Center for Cybersecurity

Brooklyn, NY, USA

Graduate Research Assistant

Sept. 2021 – Jan. 2024

- Developed game-theoretic, neuro-symbolic, and adaptive-learning methods for autonomous driving, navigation security, and automated penetration testing. Publications include IEEE TIFS (2025), IEEE MILCOM (2024), and IEEE ICRA (2023).

## TEACHING EXPERIENCE

**Teaching Assistant**, ECE-GY 5213 Introduction to System Engineering, New York University   Spring 2024

## TECHNICAL EXPERTISE AND EXPERIMENTAL SYSTEMS

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- **Statistical inference and learning:** Likelihood and conditional-posterior estimation; Monte Carlo marginalization over latent position and orientation; multi-observation Bayesian fusion; calibration with proper scoring rules and empirical coverage; U-Net, attention, and Transformer models.
- **Sequential decision-making and optimization:** Physics-informed and neuro-symbolic reinforcement learning; PPO and recurrent actor-critic policies under partial observability; bandit learning; online meta-adaptation; distributed online convex optimization.
- **Wireless digital-twin simulation and calibration:** Sionna RT scene and trajectory design; TX/RX pose and mobility simulation; geometry, material, and pose variation; measured antenna-pattern integration; stochastic RF-chain and noise modeling; Wireless InSite simulation.
- **Robotics and multimodal sensing:** System architecture and experiment leadership for *MobiFR3*; ROS 2 integration with TurtleBot4; Jackal UGV integration in progress; synchronized RF, odometry, LiDAR, and RGB observations for localization, SLAM, and navigation.
- **RF and software-defined radio systems:** Experiment design and supervision for a 10-GHz FR3 RFSoc/Pi-Radio software-defined radio (SDR) channel sounder with Vivaldi antennas, D48 pan-tilt actuation, and linear-track measurements; hands-on TI AWR1642/AWR6843 point-cloud acquisition and filtering for UAV trajectory extraction.
- **Scientific computing:** Python, MATLAB, R, Linux and shell scripting, LaTeX, Kubernetes; SUMO and CARLA for mobility and autonomous-driving simulation.

## INDUSTRY EXPERIENCE

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### JD.com, Inc.

Algorithm Engineer Intern

Beijing, China

Jan. 2021 – Feb. 2021

- Developed CenterNet/PIoU-based rotated-object detection and supported integration and on-site validation in a manufacturing laboratory.

## HONORS AND AWARDS

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- Ernst Weber Fellowship, NYU Tandon School of Engineering, 2023
- Tandon Research Excellence Exhibit, *AI-Driven Interactive Safe Autonomous Driving*, 2022
- School of Engineering Fellowship, NYU Tandon School of Engineering, 2022
- China National University Student Innovation and Entrepreneurship Training Program, 2017–2019

## PROFESSIONAL SERVICE

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**Journal reviewer:** IEEE Transactions on Signal Processing; IEEE Transactions on Communications; IEEE Wireless Communications Letters; IEEE Transactions on Vehicular Technology; IEEE Vehicular Technology Magazine; IEEE Control Systems Letters.

**Conference reviewer:** IEEE ICC 2026; IEEE INFOCOM 2025; IEEE GLOBECOM 2025; IEEE CDC 2025; IEEE ICRA 2024; ACC 2024.

## REFERENCES

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Prof. Sundeep Rangan, Director, NYU WIRELESS; Professor, NYU Tandon

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Prof. Ludovic Righetti, Professor, NYU Tandon; International Chair, ANITI

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Prof. JohnRoss Rizzo, MD, Associate Professor, NYU Grossman School of Medicine

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Prof. Marco Mezzavilla, Associate Professor, Politecnico di Milano

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Prof. Tao Li, Assistant Professor, City University of Hong Kong

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